OO-D-738C
February 10, 1978
SUPERSEDING
OO-D-738B
October 27, 1969

FEDERAL SPECIFICATION

DRY CLEANING PRESSES AND PUFF IRONS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers commercial type dry cleaning presses and puff irons for cotton, wool or silk.

1.2 Classification. The dry cleaning presses, and puff irons shall be of the following types and styles as specified (see 6.2).

Type I - Press, utility

Style 1 - Shoulder puff iron
Style 2 - Steam iron with attachment

Type II - Press, trouser

Type III - Press, mushroom

Type IV - Puff iron individual, egg

Type V - Puff iron individual, sleeve

Type VI - Puff iron individual, mushroom

Type VII - Puff iron individual, shoulder

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 3510

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Federal Specifications:

QQ-N-281	-	Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-288	-	Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.

Federal Standard:

FED-STD-123	-	Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-3153	-	Laundry and Drycleaning Machinery and Equipment (for Fixed Installations), Preparation for Delivery of.

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society of Mechanical Engineers (ASME)

Boiler and Pressure Vessel Code, Section VIII, Pressure Vessels

(Application for copies of the ASME Codes should be addressed to the Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

American Society for Testing and Materials (ASTM)

A 167 - Stainless and Heat Resisting Chromium-Nickel, Steel, Plate, Sheet and Strip

A 175 - Stainless and Heat Resisting Chromium Steel, Plate, Sheet and Strip

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. Each item of equipment covered by this specification shall be the manufacturer's current standard product except for changes necessary to comply with the requirements of this specification.

3.2 ASME Code and certification. The chest, chamber and cylinder of the dry cleaning presses shall be in accordance with ASME Boiler and Pressure Vessel Code. Acceptable evidence of meeting the requirements of ASME shall be a written certificate stating that each chest, chamber and cylinder has been marked with the ASME official U symbol and that they comply with the provisions of the ASME Code.

3.3 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for dry cleaning presses and puff irons provided the completed items comply with all provisions of this specification (see 6.5).

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3.3.1 Corrosion-resistant metal. Corrosion-resistant metal shall conform to: ASTM Specifications A 167 or A 176 as applicable, nickel-copper alloy of QQ-N-281 or composition A of QQ-N-288.

3.3.2 Materials for fasteners. Rivets, bolts, screws, nuts and washers shall be made of steel, brass or corrosion-resistant metal.

3.3.3 Pipe, tube, fittings and valves. Pipe, tube, fittings, and valves used for water, air, and steam lines shall comply with standard commercial items designed for the intended use.

3.4 Design and construction. The dimensions of the pressing surfaces for the dry cleaning presses and the puff irons shall be as specified in tables I and II. Exposed moving parts shall be fully inclosed or guarded for the protection of personnel. Means shall be provided for anchoring components to the floor or table top, as applicable. Assemblies of pipes, tubes, fittings, cylinders, chests and chamber except spray chambers using steam shall be designed for a working pressure up to 125 pounds per square inch (862 kPa) gauge, and shall be capable of withstanding a hydrostatic pressure in accordance with ASME Pressure Vessel Code when tested as specified in 4.3.1. Air lines and cylinders shall be designed to withstand a minimum of 80 psi (552 kPa) gauge when tested as specified in 4.3.2. The completed presses and puff irons shall conform to the following requirements when tested as specified in 4.3.3.

(a) The assembly of the components shall have complete and proper interrelations and adjustment.

(b) Starting, stopping, and control devices shall operate smoothly and positively.

(c) Pressure heads are in proper adjustment.

(d) No structural failures.

(e) No leaks.

Where dissimilar metals are fastened, the fasteners shall be corrosion-resistant metal. Each steam supply and return line shall be equipped with the necessary valves and condensate trap with bypass assembly.

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TABLE I. Pressing surface dimensions

	Type I	Type II	Type III
Length, inches (mm)	44(1118) minimum	43(1092) minimum	19(483) minimum
Width, inches (mm) at widest point	12-3/4(324) maximum	19-1/4(489)	9-1/4(235), oval
Width, inches (mm) at narrowest point	8-1/4(210) minimum	12(305)	Not applicable
Height, inches (mm) above floor	40-1/2 ± 1 (1030 ± 25.4)	40-1/2 ± 1 (1030 ± 25.4)	40-1/2 ± 1 (1030 ± 25.4)

TABLE II. Puff iron dimensions

Type IV	Type V	Type VI	Type VII
1-3/4 to 2-3/8 by 3-1/4 inches (45 to 60 by 83 mm)	2-1/2 by 14 to 22 inches (64 by 356 to 559 mm)	7-3/4 by 12 inches (197 by 305 mm)	4 by 6-3/4 inches (102 by 171 mm)

3.4.1 Presses, dry cleaning. The dry cleaning presses shall conform to the construction details specified herein. The type I, style 1 press shall be furnished with a shoulder puff iron. The type I, style 2 press shall be furnished with a steam iron with attachment.

3.4.1.1 Frame. The frame shall support the head, buck, table and operating mechanism. The frame shall be provided with removable panels to enclose the operating mechanism.

3.4.1.2 Buck. The buck shall be a cast or fabricated ferrous metal, convex in shape with a pressure chamber for heating and a spray chamber for moistening. The distance between the widest point on the buck and the backboards shall be not less than 6 inches (152 mm). A pedal operated buck steam valve shall be provided to deliver steam to the spray chamber and the valve shall be located for uniform distribution of steam over the pressing surface. The buck shall be provided with an air-vacuum drying valve.

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3.4.1.3 Head. The head shall be fabricated from steel, concave in shape to conform to the buck, with a pressure chamber for heating and a spray chamber for moistening. The surface of the head spray chamber shall be polished and perforated. The head shall be provided with a removable, knurled, nickel-copper or copper alloy metal press plate, that contacts the garments being processed to prevent shine or gloss. The removable plate shall be provided with a diffuser plate for insertion between this plate and the inner perforated surface of the head. The head shall overlap the buck not less than 1/2 inch (13 mm) all around and shall be connected to the steam supply and return lines by a flexible metallic hose. A head steam valve shall be provided to deliver steam to the spray chamber.

3.4.1.4 Head carrying member. The head carrying member shall be attached to a swinging or rocking member provided with a bearing at the fulcrum point. The head carrying member shall be arranged to raise the head to a position that will not interfere with the playing of work on the buck. The front of the head shall be not less than 14-1/2 inches (368 mm) from the buck in full open position when tested as specified in 4.3.3. Also an arrangement shall be provided for variable pressure by use of a handwheel mounted on the head carrying member or on the front panel of the frame when tested as specified in 4.4.3. The head shall be counterbalanced and provided with a shock absorber to permit checking of the head to a smooth, silent stop, but not interfere with rapid opening of the head.

3.4.1.5 Table. A rigid and neatly furnished wood or metal table with a backboard shall be provided to support the work and to prevent contact of work with the levers, piping and frame of the machine. The top of the table shall be 30 $\pm$ 1 inch (762 $\pm$ 25.4 mm) from the floor.

3.4.1.6 Bearing. All contact metal surfaces subject to movement during normal operation, shall be provided with bronze bushings or ball or roller bearings. Bronze bushings, when used, shall be readily replaceable. All bronze bushings shall be of the oilless or oil-impregnated type.

3.4.1.7 Pneumatic controls. The downward movement of the head shall be controlled by pneumatic controls. These controls shall bring the head to within 1/2 inch (13 mm) or less of the padded buck without locking and when released the head shall automatically return to the open position when tested as specified in 4.3.3. Locking shall be accomplished by an additional valve only after the head is 1/2 inch (13 mm) or less from the padded buck. Pressure shall be capable of finishing military coats and jackets without damage to aluminum or brass buttons with metal shanks when tested as specified in 4.3.4.

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3.4.2 Puff irons. The puff irons shall consist of a pressure chamber for heating and a spray chamber for moistening. The surfaces of the spray chamber shall be perforated uniformly over the spraying area. The irons shall be provided with stands, which shall be attached to an enameled carbon steel or corrosion-resisting steel tabletop, to provide support for the unit and the garments being serviced. The tabletop shall be 30 ± 1 inch (762 ± 25.4 mm) above the floor. The puff iron heads shall be fabricated from steel, aluminum, or nickel-copper. Steel heads shall be chromium or nickel plated. All heads shall be highly polished. A pedal operated steam valve shall be provided to deliver steam to the spray chamber of each puff iron when tested as specified in 4.3.3 by depressing the pedal. Steam shall be distributed evenly through the perforations.

3.4.3 Lubrication. Lubrication fittings shall be provided and accessibly located for force type systems.

3.4.4 Finish. The presses and puff irons shall be finished in accordance with standard commercial practice.

3.5 Marking for identification. The press and puff irons shall be marked for identification in accordance with the manufacturer's standard commercial practice and shall include manufacturer's name, date of manufacture, and model number.

3.6 Contract data requirements. The contract data requirements shall be as specified (see 6.2 and 6.4).

3.7 Government-furnished property. Military coats and jackets shall be government-furnished property (see 6.2 and 6.3).

3.8 Workmanship. The dry cleaning presses and puff irons shall be free from defects such as fractures, splits, punctures, tears, dents, creases, deteriorations, or malfunctions. There shall be no sharp edges, slivers, burrs, projections, or other defects.

3.8.1 Application of finish. The finish applied to the end item shall be continuous, smooth, adherent, without discoloration or foreign material imbedded, and contain no sags, runs, drips, creeps, laps, bubbles, streaks, wrinkles, blisters, cracks, scratches, pores, pits, lumps, flux, or orange peel or area of no film. No rust, rough grinds, or toolmarks shall show through the coatings.

3.8.2 Welding. The surfaces of parts to be welded shall be free from oxide, scale, paint, grease, and other foreign matter. Welds shall be continuous, sound, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux (when flux is used) shall be removed from the finished welds.

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3.8.3 Soldering. Soldering shall be complete and adherent with all flux and residue removed and shall contain no pin holes.

3.8.4 Threaded fasteners. Threaded fasteners shall not be broken, stripped, fractured, or loose.

3.8.5 Rivets. Rivets shall be driven to completely fill the holes, with full size heads neatly finished and in full contact with the surface of the riveted members.

3.8.6 Electrical wiring. Wiring shall not be cut, abraded, or have insulation stripped, and shall be joined at terminals. Wiring shall have adequate slack to provide relief strain.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except as otherwise specified herein, sampling shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications and standards, unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 End item inspection. The inspection lot shall consist of all dry cleaning presses or all puff irons of the same description offered for acceptance at one time. The sample unit shall be one completely fabricated dry cleaning press or puff iron, as applicable.

4.2.2.1 Visual examination. Examination shall be made of the end item for defects in table III. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE III. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Type of finish not as specified, or not in accordance with the manufacturers standard commercial practice		X
	Area of no film	X	
	Not continuous, smooth and adherent	X	
	Discoloration or foreign material, embedded in finish		X
	Sags, runs, drips, creeps, laps, bubbles, streaks, wrinkles, blisters, cracks, scratches, pores, pits, lumps, flux or orange peel		X
	Rust, rough grinds or toolmarks showing through the coating	X	
Design and construction and workmanship (applicable to all components and assemblies)	Type and style, as applicable, not as specified	X	
	Part missing	X	
	Any component fractured, split, punctured, dented, torn, creased, or malformed	X	
	Any burr, sharp edge or sliver	X	
	Exposed moving parts not fully enclosed or guarded	X	
	Provision for anchoring to floor or table, as applicable, not provided	X	
Frame	Not supporting the head buck, table and operating mechanism	X	
	Removable panels to enclose the operating mechanism not provided	X	
	Components not accessible for operation and maintenance	X	
	Part not in proper alignment	X	
	Bearings not readily replaceable	X	
Lubrication fittings (where applicable)	Missing	X	
	Not accessible for servicing	X	

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TABLE III. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Bolts, nuts, screws, studs and other types of threaded components	Broken, stripped, fractured or not tight	X	
Welding and brazing (where applicable)	Missing, incomplete, burn holes, fractured, or not fused, slag inclusion, undercut, not smooth and uniform, or deposit of scale and flux not removed	X	
Soldering (where applicable)	Missing, incomplete or not adherent	X	
	Not clean, (flux or residue not removed), or contains pinholes		X
Electrical assembly	Wiring cut, abraded, or not properly installed or joined		X
	Adequate slack not provided for wiring to relieve strain		X
Marking for identification	Missing, incomplete, not legible, or stamped		X

4.2.2.2 Dimensional examination. Inspection shall be made of the end item to determine compliance with dimensions specified. Any dimension not within the specified tolerance shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.2.3 Testing of the end item. A random sample from the first production lot shall be tested as specified in 4.3.1, 4.3.2, 4.3.3 and 4.3.4. Any nonconformance shall be cause for rejection of the lot. Each unit of production shall be tested as specified in 4.3.3. Any nonconformance shall be cause for rejection of the unit.

4.2.3 Code compliance. Proof of compliance with ASME requirements, as specified in 3.2, shall be available to the Government representative.

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4.2.4 Inspection of preparation for delivery. Preservation-packaging, packing, and marking shall be inspected in accordance with the quality assurance provisions of MIL-L-3153.

4.3 Tests.

4.3.1 Hydrostatic pressure tests. The chests and chambers using steam shall be tested with a hydrostatic pressure in accordance with the provisions of the ASME Unfired Pressure Vessel Code for the working pressure specified in 3.4. All assemblies of pipes, fittings, coils, and cylinders using steam shall be tested with the same hydrostatic pressure. Nonconformance to 3.4 shall constitute failure of this test.

4.3.2 Air pressure tests. All air lines and cylinders shall be tested with 85 ± 5 psi (586 ± 35 kPa) of air pressure. The test pressure shall be maintained for 10 minutes to determine that no leaks are present. Nonconformance to 3.4 shall constitute failure of this test.

4.3.3 Operational tests. Each press and puff iron shall be connected to air and steam (as applicable) and operationally tested for at least 10 minutes to determine compliance with the requirements of 3.4 a, b, c, d and e and 3.4.1.4, 3.4.1.7, and 3.4.2.

4.3.4 Functional tests. The press or puff iron, as applicable, shall be functionally tested by pressing GFP military coats and jackets without damage to aluminum or brass buttons and shanks (see 3.4.1.7). During this test the unit shall also be checked for 4.3.3. Nonconformance shall constitute failure of the test.

5. PACKAGING

5.1 Preservation-packaging and packing. Preservation-packaging and packing for level A, B, or C shall be in accordance with MIL-L-3153 (see 6.2).

5.2 Marking. Marking shall be as specified (see 6.2).

5.2.1 Civil agencies. All markings shall be in accordance with FED-STD-123.

5.2.2 Military requirements. All marking shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Dry cleaning presses and puff irons covered by this specification are intended for use in fixed dry cleaning plants.

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6.2 Ordering data. Purchasers should exercise options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and style of dry cleaning presses or type of puff iron required (see 1.2).
(c) Contract data requirements (see 3.6).
(d) Source and delivery of GFP (see 3.7).
(e) Selection of applicable levels of preservation-packaging and packing (see 5.1).
(f) Marking required (see 5.2).

6.3 Government-furnished property. The contracting officer should arrange to furnish the property specified (see 3.7 and 6.2).

6.4 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification shall be specifically called for in the contract in accordance with the applicable regulations of the procuring agency (see 3.6 and 6.2).

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification (see 3.3).

6.6 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value in US customary units shall be controlling.

6.7 Type designation and deletion data. Types, classes, styles, and designs indicated in OO-D-738B (October 27, 1969) were completely revised to reflect options currently required.

Custodians:

Army - GL
Navy - YD
Air Force - 99

Review activities:

Army - MD
Air Force - 45
DLA - GS

User activity:

Army - CE

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

GSA-FSS
HEW-NIH
VA-DMS

Project No. 3510-0164

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